Logic Exercises 3

1. Which of the following five formulas are in disjunctive normal form (DNF), and whichones are in conjunctive normal form (CNF)?

a) $p \wedge \neg(q \vee r)$

Neither CNF nor DNF: $\wedge$ □ it can't be DNF, $\neg(q \vee r)$ □ can't be CNF

b) $p \vee q \vee \neg r$

$p \vee q \vee r$ is DNF, $(p \vee q \vee r) \wedge \top$ □ CNF

c) $p \wedge q \wedge \neg r$

$p \wedge q \wedge r$ □ CNF, $(p \wedge q \wedge r) \vee \bot$ □ DNF

d) $p \vee (q \wedge \neg r)$

DNF (not CNF because $\wedge$)

e) $p \wedge (q \vee \neg r)$

CNF (not DNF because $\vee$)

2. Construct formulas in DNF and in CNF that are semantically equivalent to theformula ϕ, based on its truth table:

p	q	r	ϕ
T	T	T	T
T	T	F	F
T	F	T	F
T	F	F	F
F	T	T	T
F	T	F	F
F	F	T	T
F	F	F	F

DNF:

Row 1: $p \wedge q \wedge r$

Row 5: $\neg p \wedge q \wedge r$

Row 7: $\neg p \wedge \neg q \wedge r$

□ $(p \wedge q \wedge r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge r)$

CNF:

Row 2: $\neg p \vee \neg q \vee r = \neg(p \vee q \vee \neg r)$

Row 3: $\neg p \vee q \vee \neg r$

Row 4: $\neg p \vee q \vee r$

Row 6: $p \vee \neg q \vee r$

Row 8: $p \vee q \vee r$

□ $(\neg p \vee \neg q \vee r) \wedge (\neg p \vee q \vee \neg r) \wedge (\neg p \vee q \vee r) \wedge (p \vee \neg q \vee r) \wedge (p \vee q \vee r)$

3. (a) Express the formula $p \to q$ using only the Sheffer stroke | (and p and q).

NEED TO KNOW:

$p \to q \equiv \neg p \vee q$

$p \mid p \equiv \neg p$

$p \mid q \equiv \neg(p \wedge q) \equiv \neg p \vee \neg q$

$p \to q \equiv \neg p \vee q \equiv \neg p \vee \neg(\neg q) \equiv \neg p \vee \neg (q \mid q) \equiv p \mid (q \mid q)$

(b) Build a tautology and a contradiction using only the Sheffer stroke (and a propositional variable).

Tautology: $p \vee \neg p \equiv \neg p \vee p \equiv p \to p \equiv p \mid (p \mid p)$

Contradiction: $p \wedge \neg p \equiv \neg(\neg p \vee p) \equiv \neg (p \mid (p \mid p)) \equiv (p \mid (p \mid p)) \mid (p \mid (p \mid p))$

4. (a) Show that $\{\neg, \to\}$ is an adequate system of connectives (meaning that every truth table can be expressed using a formula built from only these connectives and some propositional variables).

Need to prove that $\neg, \vee, \wedge$ can be expressed

$\neg$: already in $\{\neg, \to\}$

$\vee$: $p \to q \equiv \neg p \vee q$

- $p \to \neg q \equiv \neg\neg p \vee q = p \vee q$

From $\neg$ and $\to$ we can express $\{\neg, \vee\}$ which is an adequate, therefore $\{\neg, \to\}$ is adequate

And: $p \to q \equiv \neg p \vee q$

- $p \to \neg q \equiv \neg p \vee \neg q = \neg(p \wedge q)$
- $\neg(p \to \neg q) \equiv \neg\neg p \wedge q \equiv p \wedge q$

From $\neg$ and $\to$ we can express $\{\neg$, and$\}$ which is an adequate, therefore $\{\neg, \to\}$ is adequate

(b) Argue that $\{\wedge, \vee, \to\}$ is not an adequate system of connectives.

p	$p \wedge p$	$p \vee p$	$p \to p$	$? \equiv \neg p$
T	T	T	T	F
F	F	F	T	T

$\neg p$ doesn't exist as $p \wedge p$ and $p \vee p$ are the same as p and $p \to p$ is a tautology so can't express something that gives anything other than <T,F> or <T,T>

(c) Is $\{\neg, \leftrightarrow\}$ an adequate system of connectives?

p	q	$\neg p$	$\neg q$	$p \leftrightarrow q$	$p \to q \equiv ?$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	F	T
F	F	T	T	T	T

Each have an even number of true values so any combination will always give an even number of true values hence $\{\neg, \leftrightarrow\}$ isn't adequate as we can't express any formula that has an odd number of true values e.g. $p \rightarrow q$

5. (a) Determine a CNF of the formula $(\neg q \wedge r) \rightarrow (p \wedge \neg q)$ by applying the algorithm CNF.

$(\neg q \wedge r) \rightarrow (p \wedge \neg q) \rightsquigarrow \neg(\neg q \wedge r) \vee (p \wedge \neg q)$ (IMP-FREE)

$\rightsquigarrow (\neg\neg q \vee \neg r) \vee (p \wedge \neg q)$ (NEGATION)

$\rightsquigarrow (q \vee \neg r) \vee (p \wedge \neg q)$ (NEGATION)

$\rightsquigarrow ((q \vee \neg r) \vee p) \wedge ((q \vee \neg r) \vee \neg q)$ (DISTRIBUTE)

$\rightsquigarrow (q \vee \neg r \vee p) \wedge (q \vee \neg r \vee \neg q)$ (DISTRIBUTE)

(b) Do the same for the formula $\neg\neg p \rightarrow \neg(p \rightarrow \neg p)$.

$\rightsquigarrow \neg\neg p \rightarrow \neg(\neg p \vee \neg p)$ (IMP-FREE)

$\rightsquigarrow \neg\neg p \vee \neg(\neg p \vee \neg p)$ (IMP-FREE)

$\rightsquigarrow \neg p \vee (\neg\neg p \wedge \neg\neg p)$ (NEGATION)

$\rightsquigarrow \neg p \vee (p \wedge p)$ (NEGATION)

$\rightsquigarrow (\neg p \vee p) \wedge (\neg p \vee p)$ (DISTRIBUTE)

(c) Apply the criterion for determining whether a CNF is a tautology (which wastreated in the lecture) to the CNFs you constructed in (a) and (b).

CNF is $\Psi_1 \wedge \Psi_2 \wedge \ldots \wedge \Psi_n$

All of Ψ_i must be tautologies

Ψ_i is a tautology if it contains p and also $\neg p$ for any variable p

(a) Is not a tautology
(b) Is a tautology

(d) Formulate an analogous criterion to determine whether a DNF is a contradiction.

DNF is $\Psi_1 \vee \Psi_2 \vee \ldots \vee \Psi_n$
It is a contradiction if all Ψ_i are contradictions
Ψ_i is a contradiction if it contains both p and $\neg p$ for any variable p

6. Apply the DPLL procedure to the following CNFs, to check if they are satisfiable.

(a) $p \wedge \neg p$

* p is a unit clause so we replace all p's with a $\top$

$\top$ or $\neg\top \equiv \top$ and $\equiv \bot$

* Therefore ϕ is not satisfiable

(b) $p \vee \neg p$

* No unit clause as there are no conjunctions

* No pure variables

* Choose p and replace all p's by $\top$

$\top$ or $\neg\top \equiv \top$ or $\bot \equiv \top$

* Therefore ϕ is satisfiable, $p = \top$ makes $\phi = \top$

(c) $(p \vee q) \wedge (\neg p \vee q) \wedge (p \vee \neg q)$

* No unit clauses

* No pure variables

* Choose p and replace all p's with $\top$

$(\top \vee q) \wedge (\neg\top \vee q) \wedge (\top \vee \neg q) \equiv \top \wedge (\bot \vee q) \wedge \top \equiv \bot \vee q \equiv q$

* q is a unit clause; replace q by $\top$

$q \equiv \top$

* Therefore ϕ is satisfiable, $p = \top$ and $q = \top$ makes $\phi = \top$

(d) $(p \vee q \vee \neg r) \wedge (\neg p \vee q \vee \neg r) \wedge (p \vee \neg q \vee \neg r) \wedge r$

* r is a unit clause so we replace all r's with a $\top$

$(p \vee q \vee \bot) \wedge (\neg p \vee q \vee \bot) \wedge (p \vee \neg q \vee \bot) \wedge \top \equiv (p \vee q) \wedge (\neg p \vee q) \wedge (p \vee \neg q)$

* No pure variables

* Choose p and replace all p's with $\top$

$(\top \vee q) \wedge (\bot \vee q) \wedge (\top \vee \neg q) \equiv \top \wedge q \wedge \top \equiv q$

* q is a unit clause, replace all q's with $\top$

$q \equiv \top$

* Therefore ϕ is satisfiable, $q \equiv \top$, $p = \top$ and $r = \top$ makes $\phi = \top$

(e) $(p \vee q) \wedge (\neg p \vee q) \wedge (p \vee \neg q) \wedge (\neg p \vee \neg q)$

* No unit clauses

* No pure variables

* Choose p and replace all p's with $\top$

$(\top \vee q) \wedge (\neg\top \vee q) \wedge (\top \vee \neg q) \wedge (\neg\top \vee \neg q) \equiv \top \wedge q \wedge \top \wedge \neg q \equiv q \wedge \neg q$

* q is a unit clause, replace all q's with $\top$

$\top \wedge \neg\top \equiv \top \wedge \bot \equiv \bot$

* Therefore not satisfiable

* Choose p and replace all p's with $\bot$

$(\bot \vee q) \wedge (\bot \vee q) \wedge (\top \vee \neg q) \wedge (\top \vee \neg q) \equiv q \wedge \top \wedge \neg q \wedge \top \equiv q \wedge \neg q$

* q is a unit clause so replace all q's by $\top$

$\top \wedge \neg\bot \equiv \top \wedge \bot \equiv \bot$

* Therefore ϕ is not satisfiable, $p = \top$ and $p = \bot$ both don't make $\phi = \top$

(f) $(p \vee q \vee r) \wedge (\neg p \vee q \vee r) \wedge (p \vee \neg q \vee r) \wedge (\neg p \vee \neg q \vee r)$

* No unit clauses

* r is a pure variable, replace all r's with $\top$

$(p \vee q \vee \top) \wedge (\neg p \vee q \vee \top) \wedge (p \vee \neg q \vee \top) \wedge (\neg p \vee \neg q \vee \top) \equiv \top \wedge \top \wedge \top \wedge \top \equiv \top$

* Therefore ϕ is satisfiable, $r = \top$ makes $\phi = \top$